

Second Hand Book Selling

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Chapter-I

Introduction

1.1 Project Overview:

The project titled “**Second Hand Book Selling**” is a full-stack web application developed to facilitate the buying and selling of second-hand academic books among college students through direct contact. The primary motivation behind this project is to reduce the financial burden on students by creating a dedicated digital marketplace for reusing academic resources in a secure and structured environment.

This system provides a user-friendly interface that allows students to create their own profile, list books for sale, search for required books, and make price offers. The application incorporates a two-way negotiation system where interested buyers can propose a price lower than the listed amount, and the seller can either accept, reject, or further negotiate the offer. Once the deal is finalized, the book can be marked as sold at the mutually agreed price, ensuring a transparent and fair transaction.

To maintain privacy and trust, the system includes a verification mechanism where only verified users can view sellers’ contact details or make offers. The admin plays a key role in verifying new users, ensuring that only genuine students participate in the platform. Additionally, once a deal is completed, both parties can exchange feedback through a built-in review system to build a reliable community of buyers and sellers.

The platform uses React JS and MUI for a responsive and clean user interface, while the backend is built with Express JS and MySQL, ensuring scalability and performance. The authentication system is secured using JWT tokens and passwords are encrypted using bcrypt. A password reset option is also available through security questions, enhancing user accessibility.

Overall, this system streamlines the traditionally unstructured and informal process of buying and selling second-hand books in colleges by providing an organized, secure, and interactive digital platform. It aims not only to make textbooks affordable but also to foster a culture of resource-sharing and sustainability among student

1.2 Objectives:

1. To provide a centralized platform for students to buy and sell second-hand books in an organized and accessible manner.
2. To ensure secure authentication using JWT tokens and encrypted passwords to protect user data and access.
3. To introduce a price negotiation system that allows buyers to make offers and sellers to accept or reject them, ensuring fair transactions.
4. To make academic books affordable by promoting peer-to-peer exchange at lower prices within the student community.
5. To implement admin verification so that only verified users can access key features like making offers or viewing contact details, ensuring safety and trust.
6. To support a post-sale review system where users can rate each other after completing a transaction to build a trustworthy environment.
7. To enable profile management and deal tracking through user dashboards where students can manage listings, offers, and personal details.
8. To promote sustainability and reuse by encouraging students to recycle academic materials instead of discarding them.

1.3 Purpose & Relevance:

The purpose of this project is to design and develop a web-based platform that facilitates the **buying and selling of second-hand academic books** among college students. The system is built to address the limitations of the current informal methods and provide a secure, structured, and efficient alternative.

The relevance of this system in the academic context is highlighted through the following key points:

- **To reduce the financial burden on students** by promoting the reuse of academic books at affordable prices, making essential resources accessible to a larger section of the student community.
- **To create a dedicated and reliable platform** where students can directly list their used books and connect with interested buyers within the campus environment.
- **To eliminate the dependency on unstructured platforms** such as WhatsApp, notice boards, or word-of-mouth communication, which often lack reach, security, and proper record-keeping.
- **To ensure user verification and privacy**, allowing only authenticated and admin-approved students to participate in transactions, thereby promoting a safe and trustworthy environment.
- **To support transparent negotiations and fair deals** through a built-in offer-making system, where the buyer can propose a price and the seller can choose to accept, reject, or counter the offer.
- **To promote sustainability and responsible academic practices** by encouraging the reuse of educational materials, thereby reducing waste and promoting eco-friendly habits.
- **To provide a user-centric system** where students can manage their profiles, track offers, review transactions, and stay updated on the status of their listings.

This project is highly relevant for students, institutions, and academic communities as it promotes both **economic efficiency and environmental awareness**, while also utilizing modern web technologies to solve a real-life campus problem

1.4 Problem Statement :

In most academic institutions, there is no formal system in place for students to exchange second-hand academic books. As a result, students often rely on **unorganized and unreliable methods** such as putting up WhatsApp statuses, sending classroom messages, or using personal contacts to buy or sell books. These methods present several problems:

- **Lack of a centralized platform:** There is no dedicated space where all book listings are available, making it difficult for students to find what they need.

- **Absence of user verification:** Informal platforms do not verify users, increasing the risk of fraud, spam, or irrelevant contacts.
- **No negotiation structure:** There is no system to facilitate offer-based pricing, which limits flexibility and often results in lost deals.
- **Privacy concerns:** In informal methods, users often have to share personal contact details publicly, compromising their privacy and safety.
- **No tracking or review mechanism:** Transactions happen without any formal record, making it difficult to resolve disputes or assess user reliability.
- **Wastage of educational resources:** Without a proper system for reuse, many books are discarded or left unused, which increases educational expenses and environmental waste.

These issues highlight the **urgent need for a verified, secure, and accessible platform** where students can easily list, discover, and exchange academic books. The “**Second Hand Book Selling**” system is designed to solve these problems by offering a digital solution tailored specifically for student use within the college ecosystem.

1.5 Scope of the Project:

The scope of the “**Second Hand Book Selling**” project is focused on providing a secure, user-friendly, and efficient web platform that allows college students to buy and sell academic books directly within their campus community. The system covers several functional and administrative aspects to ensure smooth operation and a positive user experience.

The scope includes the following key components:

- **User Registration and Authentication:** Students can create accounts and securely log in using an authentication system that uses JWT tokens and encrypted passwords to ensure data privacy.
- **Profile Management :** Users can view and edit their personal details and access features like their listed books, offers made or received, and deal history through a dedicated profile dashboard.
- **Book Listing Module :** Verified users can upload book details such as title, author, price, description, and image to offer them for sale. These listings are displayed in a central “Shop” page accessible to all verified users.

- **Offer System for Negotiation** : Buyers can make price offers on listed books, and sellers can manage these offers by accepting, rejecting, or marking a deal as sold once both parties agree on a final price.
- **Admin Verification**: A dedicated admin panel is provided to verify each user before they gain full access to platform features, especially contact details of sellers, ensuring a secure environment.
- **Transaction Management**: Once a deal is finalized, the seller can mark the book as “Sold,” and enter the agreed price. This provides clear visibility into completed transactions and maintains platform transparency.
- **Review and Feedback**: After completing a transaction, users can leave reviews to help future users identify trustworthy sellers and buyers, enhancing credibility within the system.
- **Security Features**: Password encryption, security question-based password recovery, and restricted access to sensitive data ensure the platform is secure and student-friendly.
- **Accessibility and Usability**: The system is designed using React JS and MUI for a responsive, mobile-friendly experience accessible from any device with an internet connection.

This project is currently scoped for use within a single college or institution, but the design allows for future scalability to support multiple institutions or a broader student network.

1.6 Key Features:

The “**Second Hand Book Selling**” platform includes a variety of features designed to provide a secure, efficient, and user-friendly environment for students to exchange books. The major features of the system are as follows:

i. **User Registration and Login**

A secure registration and login system with JWT-based authentication and encrypted passwords ensures only authorized access to the platform.

ii. **Admin Verification System**

Every user must be verified by the admin before gaining full access to platform functionalities like viewing seller contact information or making offers.

- iii. **Profile Management**
Each user has a dedicated profile page to manage personal information, track listed books, manage offers, and edit account details.
- iv. **Book Listing**
Verified users can list books for sale by providing essential details such as book title, author, category, description, image, and expected price.
- v. **Shop Page**
A centralized section where all active book listings are displayed. Students can browse, filter, and view details of available books.
- vi. **Offer System**
Buyers can place custom price offers for books listed by other users. Sellers can accept, reject, or counter these offers through a “Manage Offers” section.
- vii. **Deal Finalization and Sale Confirmation**
Once an offer is accepted, the seller can mark the book as “Sold” by entering the final deal amount. The book is then shown as sold with the agreed price.
- viii. **Review and Rating System**
After a successful transaction, both buyer and seller can leave reviews, building trust and transparency in future interactions.
- ix. **Security and Privacy**
Sensitive user data is protected through password encryption, and personal contact details are only visible to verified users. A password reset system via security questions is also implemented.
- x. **Responsive User Interface**
The platform uses React JS and MUI to offer a clean, modern, and mobile-friendly interface for easy navigation and usability.

These features collectively provide a reliable and transparent solution for students looking to exchange academic books securely and conveniently.

1.7 Project Category

The project “Second Hand Book Selling” falls under the category of RDBMS-based web applications. It utilizes MySQL as the backend database to manage structured data through relational tables with primary and foreign key constraints. The application is built using React JS for the frontend and Node.js with Express.js for the backend. The use of MySQL ensures data consistency, integrity, and efficient querying, aligning the project with the principles of relational database management system.

Chapter-II

Tools and requirements

This chapter outlines the tools, technologies, and system requirements used in the development of the “**Second Hand Book Selling**” project. The project is a full-stack, web-based application built using modern technologies that ensure responsive design, secure authentication, and efficient database handling.

2.1 Tools & Technologies Used :

The project is developed using the following tools and technologies :

Tool / Technology	Purpose
React JS	Frontend framework used to build user interfaces
Material UI (MUI)	React component library for UI styling and responsiveness
Node.js	Runtime environment for executing JavaScript on server-side
Express JS	Web framework for Node.js to handle backend APIs
MySQL	Relational database for storing user, book, and offer data
JWT (JSON Web Token)	Token-based authentication system

bcrypt	For password hashing and securing user credentials
Visual Studio Code	Code editor used for frontend and backend development
Postman	API testing and development tool
npm (Node Package Manager)	Package manager to install project dependencies

2.2 Hardware Requirements :

The project is designed to run on basic computer hardware typically available to students or in college labs. The minimum hardware requirements are as follows:

Component	Minimum Specification
Processor	Intel Core 2 Duo or equivalent
RAM	4 GB
Hard Disk	Minimum 2 GB free space for project files

2.3 Software Requirements :

The software environment required to develop, test, and deploy the system is listed below:

Component	Specification
Operating System	Windows 7/10/11, Linux (Ubuntu), or macOS
Code Editor	Visual Studio Code
Browser	Google Chrome / Mozilla Firefox
Node.js Version	v14 or higher
MySQL Version	5.7 or higher

Chapter-III

Problem Definition and Project Planning

This chapter presents the core problem that the project aims to address, along with a breakdown of functional and non-functional requirements, planned phases of development, and key use cases.

3.1 Problem Definition:

In most educational institutions, especially at the college level, students frequently buy and sell academic books among themselves. However, there is no dedicated or structured platform in place to facilitate this exchange. As a result, students often rely on **informal methods** such as word-of-mouth communication, WhatsApp statuses, or social media posts. These practices lead to several key challenges:

- 1. Lack of a centralized platform:**

There is no single, organized space where students can view or list books, making it difficult to discover available materials.

- 2. No system for price negotiation:**

Current methods do not support a structured way to negotiate the price of books, often resulting in deals being abandoned or misunderstood.

- 3. Limited reach and visibility:**

A WhatsApp status or class message may not reach all potential buyers, reducing the chances of successful transactions.

4. **Security and privacy concerns:**

Sharing contact information publicly on social platforms exposes students to spam, unknown users, or misuse of their data.

5. **No user verification:**

Anyone can claim to buy or sell books without being verified, which may lead to fake listings or unreliable buyers/sellers.

6. **Absence of deal tracking or feedback**

There is no system to review a transaction or track its progress, leaving room for miscommunication and lack of accountability.

Due to these problems, many students either fail to sell their books or end up purchasing new ones unnecessarily, adding to their academic expenses. This project aims to solve these issues by creating a **secure, structured, and student-friendly web application** that connects buyers and sellers directly through a verified platform, with support for listing, negotiation, deal tracking, and feedback.

3.2 Functional Requirements:

Functional requirements define the core operations that the system must perform to meet user expectations and fulfill the project's objectives. The following functionalities are essential for the successful operation of the "**Second Hand Book Selling**" platform:

1. **User Registration and Login**

- Allows users (students) to create an account and securely log in using email and password.
- JWT tokens are generated for session management.
- Passwords are securely hashed using bcrypt.

2. **Admin Verification**

- Every newly registered user must be verified by the admin before accessing certain features.
- Only verified users can make offers or view seller contact details.

3. **Book Listing**

- Verified users can add books for sale by entering details such as book title, author, price, description, category, and an optional image.

4. **View and Search Books**

- All verified users can browse listed books.
- Search functionality enables filtering by book title, author, or category.

5. **Offer System (Negotiation)**

- Buyers can submit custom price offers on available books.
- Offers are sent to the respective sellers for review.

6. **Manage Offers**

- Sellers can view all incoming offers.
- They can accept, reject, or keep offers pending.
- Once a deal is accepted, the seller can mark the book as “Sold.”

7. **Finalize Transaction**

- After accepting an offer, the seller can enter the final sale price.
- The book listing is updated to show it has been sold along with the deal amount.

8. **Review and Feedback**

- Both buyer and seller can submit a review after the deal is completed.
- Ratings help improve trust among users.

9. **Profile Management**

- Users can edit their personal information.
- Users can also view their active listings, received and sent offers, and reviews.

10. **Password Reset**

- Users can reset their password using a predefined security question and answer.

3.3 Non-Functional Requirements :

Non-functional requirements define the quality attributes, performance standards, and constraints of the system. These are equally important to ensure the application is reliable, secure, and user-friendly. The non-functional requirements for the “**Second Hand Book Selling**” system are as follows:

1. Security

- The system must implement secure user authentication using JWT (JSON Web Tokens).
- Passwords must be stored in encrypted form using bcrypt.
- Contact details should only be visible to verified users to protect user privacy.
- A password reset system using security questions must be provided.

2. Usability

- The system should have a clean, intuitive, and easy-to-use interface designed using React JS and MUI (Material UI).
- Navigation between components should be seamless, ensuring a positive user experience even for non-technical users.

3. Performance

- The application should load pages and data with minimal delay.
- Backend responses (API calls) should be optimized for speed and efficiency.
- The system should handle multiple simultaneous users without degradation in performance.

4. Scalability

- The application should be designed in a modular and scalable way so it can later support additional features or expand to multiple colleges or institutions.

5. Maintainability

- The codebase should follow proper structure and naming conventions to ensure easy maintenance.
- The system should be easy to update or upgrade as technologies evolve.

6. Compatibility

- The system should be compatible with all major web browsers including Chrome, Firefox, and Edge.
- It should work well across different devices such as desktops, laptops, tablets, and mobile phones.

7. Availability

- The application should be accessible at all times, provided the server is running.
- Users should be able to access their accounts and listings without downtime during normal usage.

3.4 Project Phases :

The development of the “**Second Hand Book Selling**” system was carried out using the **Spiral Model**, which involves repeated cycles of planning, risk analysis, development, and evaluation. Each cycle focused on specific modules and progressively built the complete system.

The major development cycles are as follows:

Cycle 1: Requirement Gathering and Risk Analysis

- Identified user needs and system requirements.
- Analyzed potential risks such as data security, privacy concerns, and user verification challenges.

Cycle 2: System Design

- Designed the system architecture, including ER diagrams and UI mockups.
- Planned database structure and frontend-backend integration.

Cycle 3: Core Module Development

- Implemented basic functionalities such as user registration, login, and book listing.
- Integrated JWT authentication and secure password handling.

Cycle 4: Feature Enhancement and Offer System

- Developed offer creation, negotiation, and management features.
- Added admin verification and deal finalization mechanisms.

Cycle 5: Testing and Risk Handling

- Conducted unit and functional testing using Postman and manual validation.
- Addressed identified risks and made improvements based on evaluation feedback.

Cycle 6: Final Review and Documentation

- Reviewed the overall system for performance, security, and usability.
- Prepared final documentation, screenshots, and report for submission.

3.5 Use Case Descriptions

This section describes the primary use cases of the “**Second Hand Book Selling**” system, highlighting the main actions performed by users and the system’s expected response. Each use case is explained briefly to demonstrate how users interact with the application.

Use Case	Description
User Registration	A student creates a new account by providing necessary details. The account remains inactive until verified by the admin.
User Login	A registered user logs into the system using a valid email and password. JWT tokens are generated for session management.
Admin Verification	The admin verifies newly registered users. Only verified users can access book listings, make offers, or view contact details.
Book Listing	A verified user can add a new book for sale by providing details such as title, author, price, and an image.
Browse Books	Users can browse all active book listings and search using filters such as title, author, or category.
Make an Offer	A verified buyer can make a custom price offer on a listed book. This offer is sent to the respective seller for consideration.
Manage Offers (Seller)	The seller can view all received offers and choose to accept, reject, or leave them pending.
Mark as Sold	After accepting an offer, the seller can mark the book as sold and enter the final deal price. The listing is updated accordingly.
Leave a Review	After the transaction is complete, both buyer and seller can submit a review and rating for each other.
Profile Management	Users can update their personal information, view their book listings, sent and received offers, and past transactions.
Password Reset	Users can reset their password by answering a predefined security question correctly.

Chapter-IV

Solution Scope

4.1 What It Solves

The “**Second Hand Book Selling**” platform is developed to address the common issues students face while buying and selling academic books. The current process, which relies on informal methods like word-of-mouth, WhatsApp statuses, and social media, is unorganized, unreliable, and lacks privacy and trust. This system provides a structured digital solution to overcome these limitations.

The project solves the following major problems:

- **Lack of a centralized system** for book exchange is addressed by offering a unified platform where all listings are visible and accessible to verified users.
- **Absence of negotiation support** is resolved through an integrated offer-making feature that allows buyers to propose prices and sellers to accept or reject them.
- **Trust and verification issues** are eliminated by introducing admin-based user verification, allowing only genuine students to access sensitive features like making offers or viewing contact details.

- **Privacy concerns** are managed by ensuring that personal information is only visible after verification, and sensitive data such as passwords are encrypted.
- **No feedback mechanism** in traditional systems is overcome with a built-in review and rating system that allows users to evaluate each other after a transaction.
- **Lack of transaction tracking** is resolved by implementing a proper workflow from listing a book to marking it as sold, including entering the final deal price.
- **Educational waste** is reduced by encouraging students to reuse books, making learning materials more accessible and sustainable.
- By solving these issues, the system brings structure, security, and convenience to the second-hand book exchange process within the college environment.

4.2 Key Users and Their Benefits

The system is designed to serve three primary types of users: **Students as Sellers**, **Students as Buyers**, and the **Administrator**. Each user role interacts with the system in different ways and receives specific benefits based on their actions and privileges.

1. Students (Sellers)

These are verified users who wish to sell their second-hand academic books.

Benefits:

- Can easily list books with details like title, author, price, and image.
- Have full control over negotiation, allowing them to accept, reject, or counter offers.
- Can mark books as sold after finalizing a deal and enter the final agreed price.
- Can manage all offers in one place through a dedicated “Manage Offers” section.
- Can receive reviews from buyers, which helps build a trusted profile for future transactions.

2. Students (Buyers)

These users browse the book listings and make offers on the books they wish to purchase.

Benefits:

- Can browse all available books and filter/search based on specific needs.
- Can make offers and negotiate prices in a transparent and structured manner.

- Can view seller contact information only after verification, ensuring safe communication.
- Can leave reviews after successful transactions, contributing to a reliable user community.
- Can track active and past offers through their profile dashboard.

3. Administrator

The admin is responsible for managing users and overseeing system operations.

Benefits:

- Has the authority to verify new users, ensuring only legitimate students gain access to sensitive features.
- Can monitor system activity and address any issues related to user misconduct or disputes.
- Helps maintain platform integrity by reviewing user actions and ensuring a safe and honest environment.

By identifying and addressing the specific needs of these key users, the system ensures a smooth, secure, and user-focused experience that promotes trust, usability, and effective book exchange among students.

4.3 Value Over Existing Methods

The “**Second Hand Book Selling**” system provides several key advantages over the traditional, informal methods used by students to exchange books. The following points highlight the most significant improvements:

1. Centralized Platform

- Unlike WhatsApp or classroom announcements, the system offers a single place where all book listings can be accessed and searched efficiently.

2. Verified User Access

- Only admin-verified users can make offers or view contact details, which ensures safety and reduces chances of spam or misuse.

3. Secure and Private

- Passwords are encrypted and personal data is protected. Contact information is only shared with verified users, maintaining privacy.

4. **Negotiation and Deal Tracking**

- The built-in offer system allows fair price negotiation, and every deal is tracked from offer to final sale.

5. **Review System**

- After a transaction, users can leave reviews and ratings, building a trustworthy environment — something missing in traditional methods.

Chapter-V

System Analysis and Design

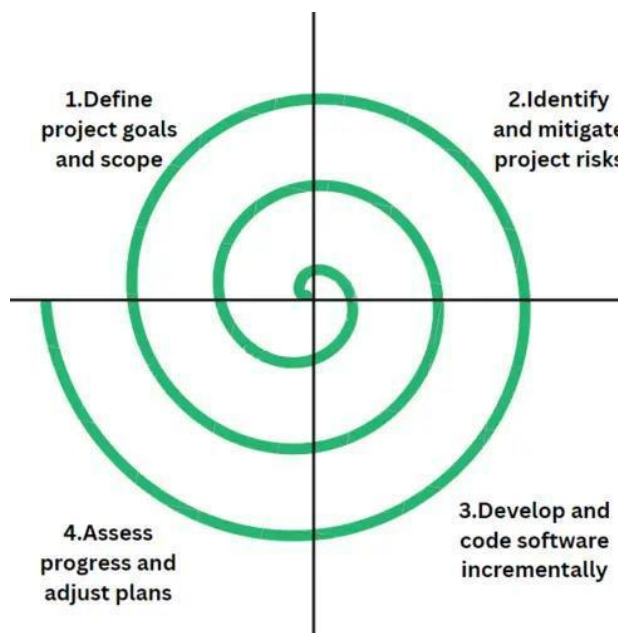
5.1 Used Software Development Life Cycle (SDLC) Model

The development of the “Second Hand Book Selling” system followed the **Spiral Model**, a risk-driven SDLC approach that blends the structured flow of the Waterfall Model with the iterative refinement of prototyping. It is well-suited for projects involving evolving requirements and risk analysis.

Each cycle in the Spiral Model consists of four key phases:

- **Planning:** Defined objectives and gathered requirements based on the challenges faced by students in second-hand book exchanges.
- **Risk Analysis:** Identified issues such as security, privacy, and user verification; addressed using JWT authentication, bcrypt encryption, and admin approval mechanisms.
- **Engineering:** Developed using React JS and MUI (frontend), Node.js and Express (backend), and MySQL (database). Modules were incrementally built and tested.
- **Evaluation:** Regular reviews and testing were conducted, and feedback was incorporated into subsequent iterations for continuous improvement.

This model ensured flexibility, progressive development, and effective risk management throughout the project.



5.2 Feasibility Study

The feasibility study evaluates whether the proposed “**Second Hand Book Selling**” system can be successfully developed and used within a college environment.

- **Technical Feasibility**

The project uses open-source and widely available technologies like React JS, Node.js, Express, and MySQL, which are well-suited for student-level development.

- **Economic Feasibility**

No paid software or hosting is required. All tools are free, making the system highly cost-effective for academic use.

- **Operational Feasibility**

The platform is easy to use for both students and administrators. Minimal technical knowledge is needed to operate the system.

- **Legal Feasibility**

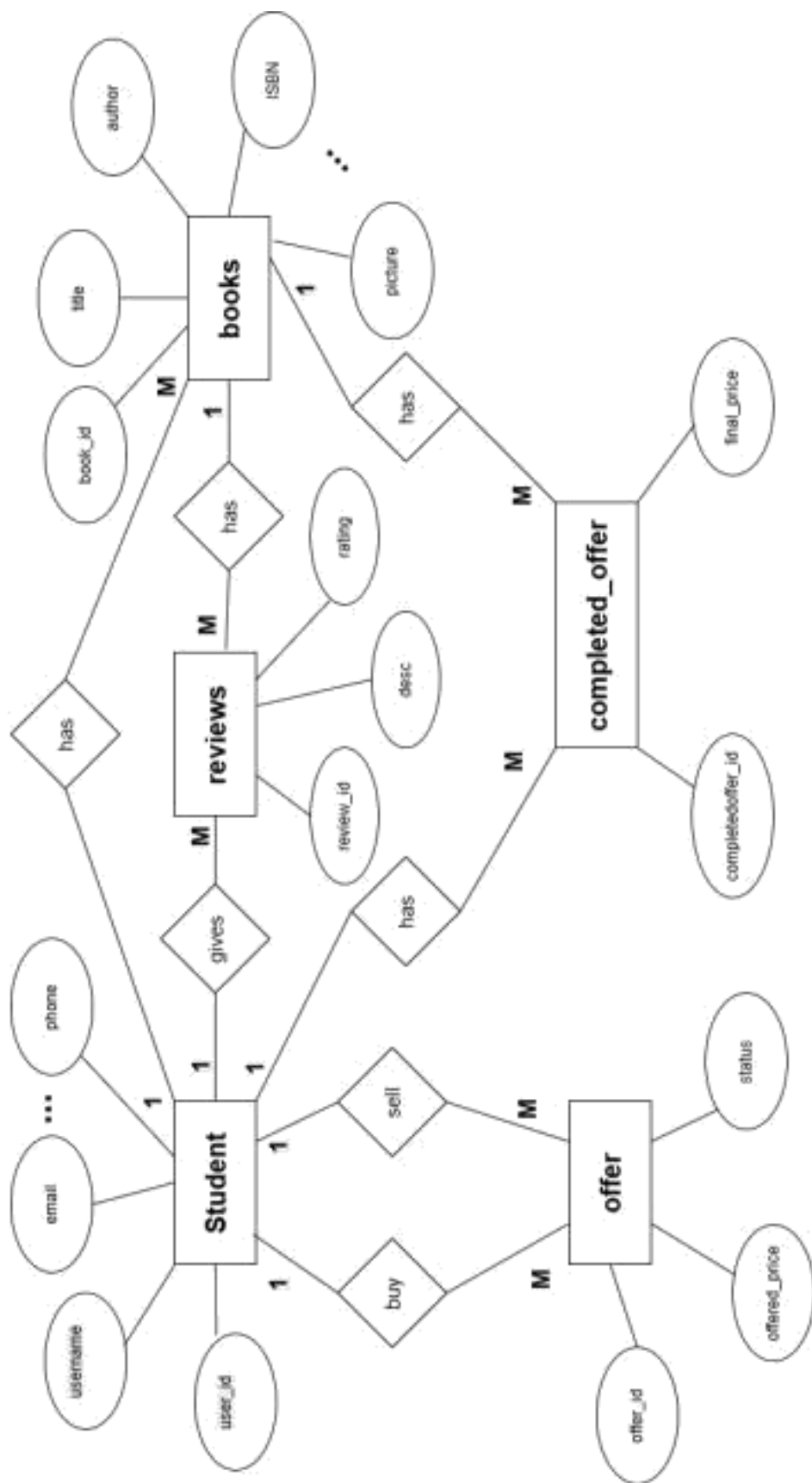
The system complies with basic privacy norms and is used only for academic purposes. No third-party content is misused.

Conclusion: The project is feasible in all aspects and can be effectively implemented within a college setting.

5.3 Entity Relationship Diagram

The following ER diagram illustrates the core entities and their relationships in the "Second Hand Book Selling" system. It visually maps how students interact with the system for buying, selling, reviewing, and completing offers for second-hand books.

- **Student:** Represents registered users. Each student can buy or sell books, give reviews, and make or receive offers.
- **Books:** Contains details like book ID, title, author, ISBN, and cover picture. One book can receive multiple reviews and be associated with multiple offers.
- **Offer:** Represents a proposal made by a buyer for a book, including the offered price and current status.
- **Reviews:** Each student can give reviews to books. Reviews are linked to both the student and the book.
- **Completed_Offer:** Represents finalized transactions between a buyer and a seller. It includes the final agreed price.



5.4 Relational Database Design

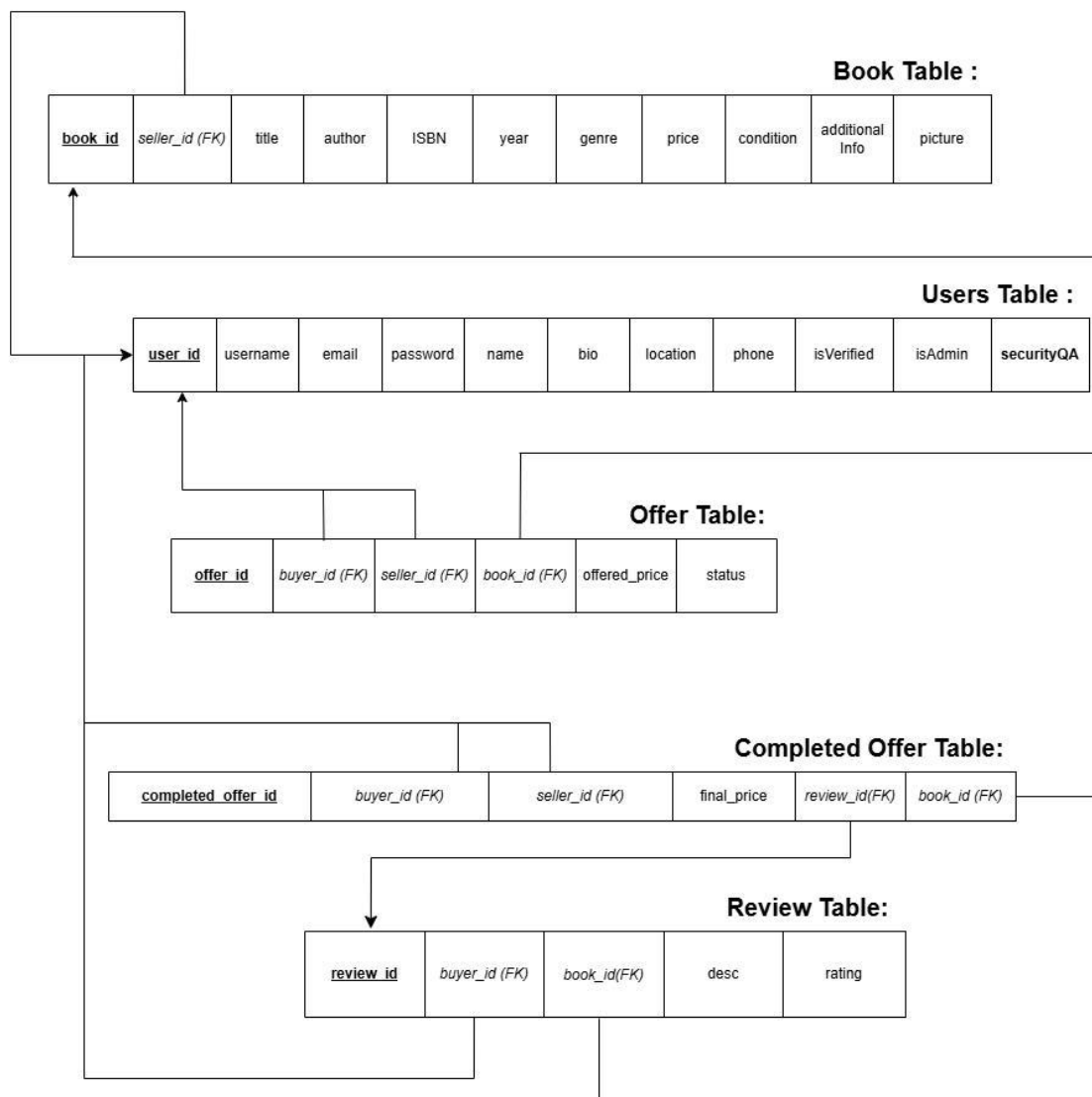


Fig 5.4 Relational Database Management System

The relational database design for the **Second Hand Book Selling System** follows a normalized schema that ensures efficient data storage, easy retrieval, and proper enforcement of relationships between different entities. The database is designed using the **MySQL** relational database management system.

5.5 Data Flow Diagrams

5.5.1 Context Level DFD (Level 0)

The **Context Level DFD** provides a high-level overview of the entire system, showing the interaction between external entities and the main system process

Level 0

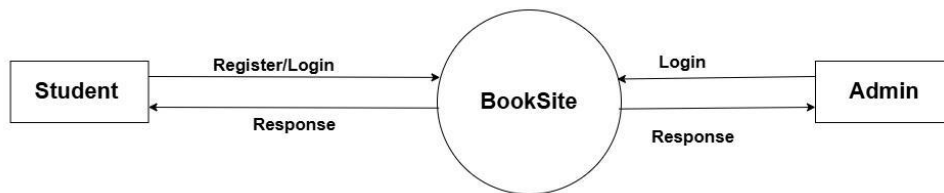
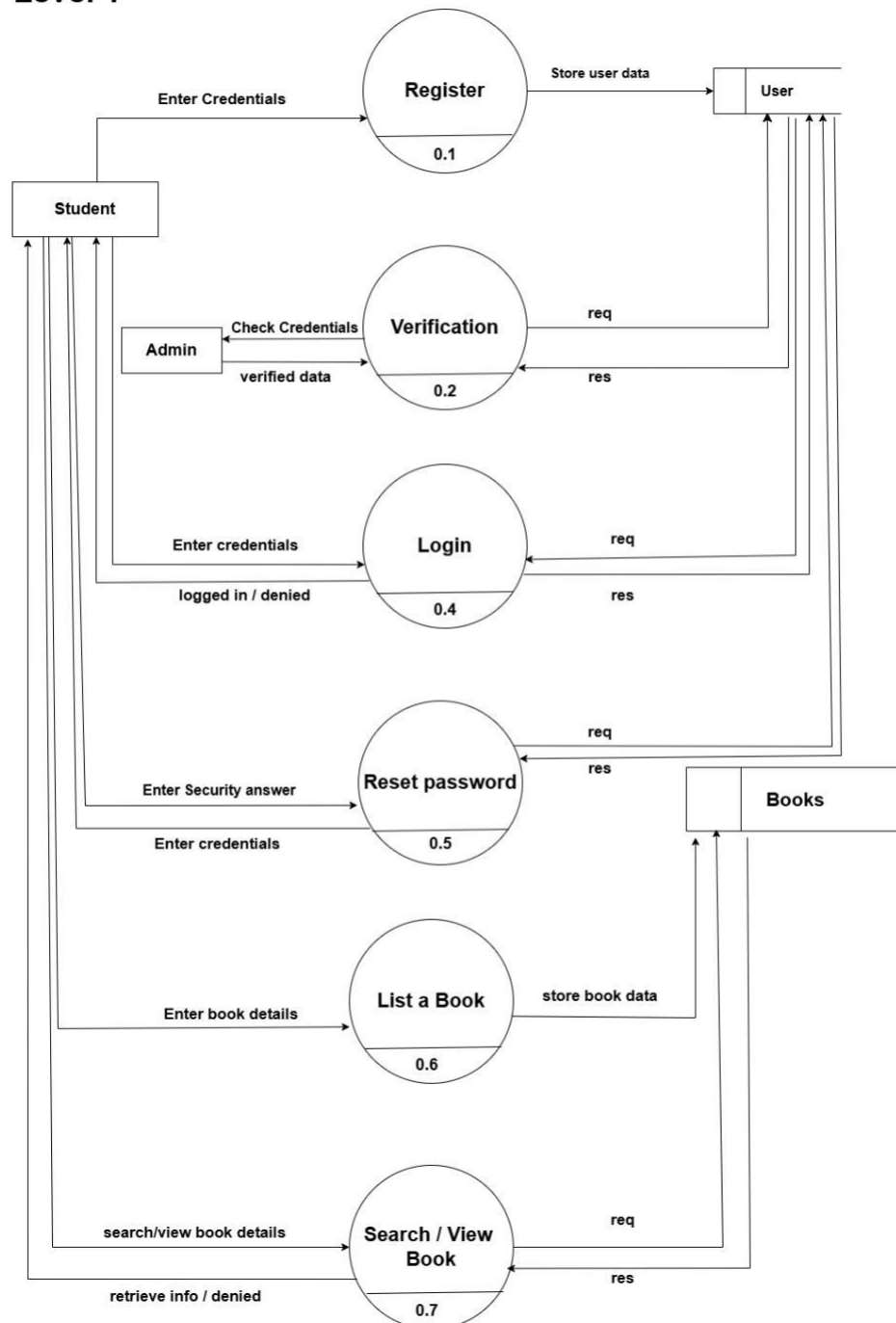


Fig. 5.5.1: Context Level DFD

5.5.2 Level 1 DFD

The **Level 1 DFD** breaks down the main functionalities of the system into subprocesses for better clarity

Level 1



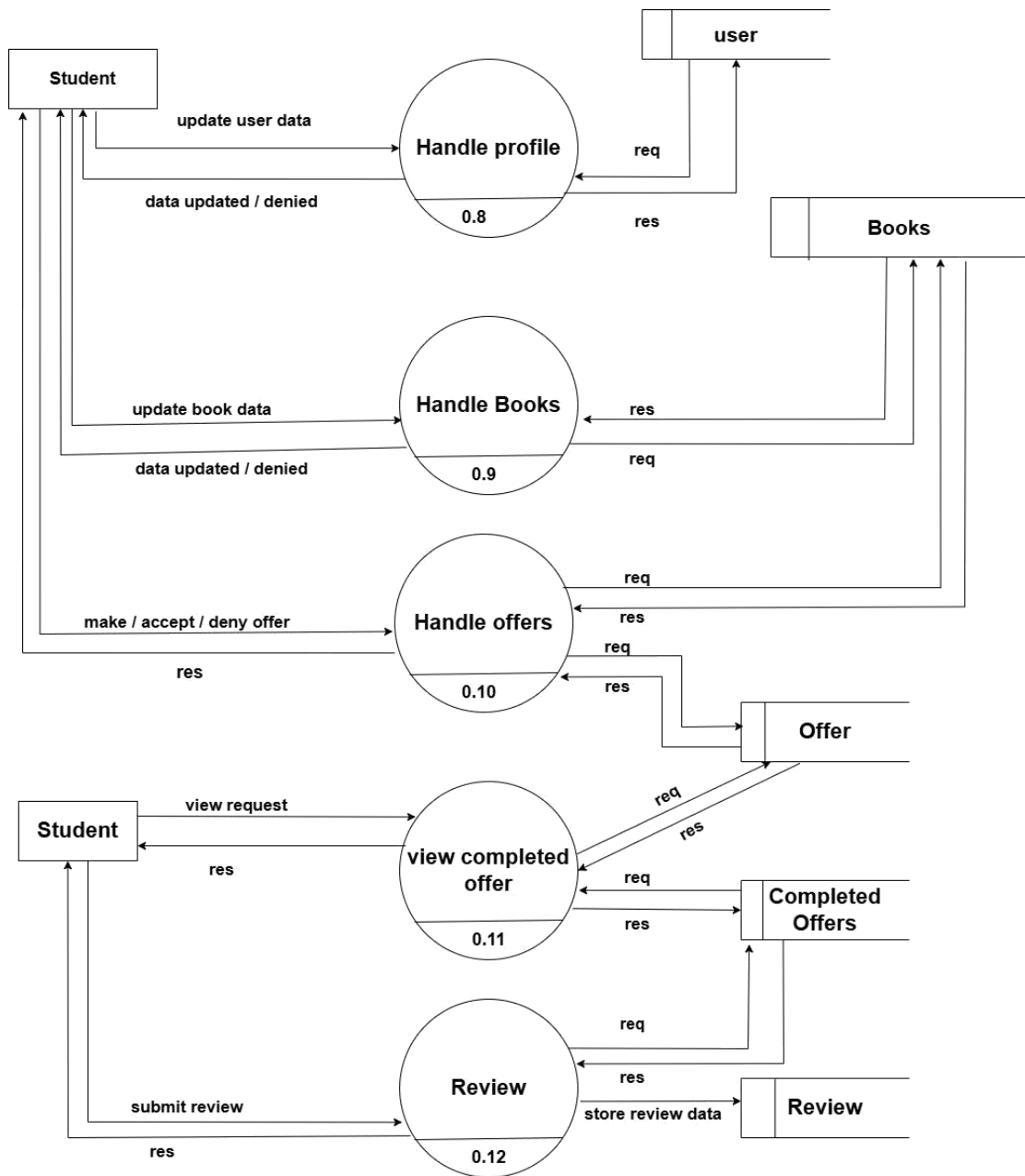


Fig 5.5.2 Level 1 DFD

5.6 Data Dictionary :

Books Table

Field Name	Data Type	Nullable	Description
id	int	No	Unique identifier for each book listing
title	varchar(255)	No	The title of the book
author	varchar(255)	No	The author(s) of the book
ISBN	varchar(255)	No	International Standard Book Number for book identification
year	int	No	The year the book was published
genre	varchar(255)	No	The category or genre of the book
price	float	No	The asking price for the book in local currency
condition	enum('New','Good','Fair','Poor')	No	The physical condition of the book
picture	text	Yes	URL or base64 encoded image of the book
quantityAvailable	int	No	Number of copies available for sale
additionalInfo	text	Yes	Additional information about the book provided by seller

createdAt	datetime	No	Timestamp when the book listing was created
updatedAt	datetime	No	Timestamp when the book listing was last updated
sellerId	int	Yes	Foreign key reference to the user who is selling the book

Users Table

Field Name	Data Type	Nullable	Description
id	int	No	Unique identifier for each user
username	varchar(255)	No	User's chosen username for login and display
email	varchar(255)	No	User's email address for notifications and account recovery
password	varchar(255)	No	Encrypted password for user authentication
name	varchar(255)	No	User's full name
bio	text	Yes	User's self-written biography or description
location	varchar(255)	Yes	User's geographical location
phone	varchar(255)	No	User's phone number for contact and verification
isVerified	tinyint(1)	Yes	Flag indicating if the user's account has been verified
isAdmin	tinyint(1)	Yes	Flag indicating if the user has administrative privileges

createdAt	datetime	No	Timestamp when the user account was created
updatedAt	datetime	No	Timestamp when the user account was last updated
securityQA	json	Yes	Security questions and answers for account recovery

Offers Table

Field Name	Data Type	Nullable	Default Value	Description
id	int	No		Unique identifier for each offer
offeredPrice	float	No		The price offered by the buyer for the book
status	enum('Pending','Accepted','Rejected','Completed')	Yes	Pending	Current status of the offer
createdAt	datetime	No		Timestamp when the offer was created
updatedAt	datetime	No		Timestamp when the offer was last updated
bookId	int	Yes		Foreign key reference to the book being offered on
sellerId	int	Yes		Foreign key reference to the user selling the book
buyerId	int	Yes		Foreign key reference to the user making the offer

Completed offers Table

Field Name	Data Type	Nullable	Description
id	int	No	Unique identifier for each completed transaction
finalPrice	float	No	The final agreed price for the book sale
completedAt	datetime	Yes	Timestamp when the transaction was finalized
createdAt	datetime	No	Timestamp when the completed offer record was created
updatedAt	datetime	No	Timestamp when the completed offer record was last updated
bookId	int	Yes	Foreign key reference to the book that was sold
sellerId	int	Yes	Foreign key reference to the user who sold the book
buyerId	int	Yes	Foreign key reference to the user who purchased the book

Reviews Table

Field Name	Data Type	Nullable	Description
id	int	No	Unique identifier for each review
rating	int	No	Numerical rating given to the book (typically 1-5)
reviewText	text	No	The written review content provided by the buyer
createdAt	datetime	No	Timestamp when the review was created
updatedAt	datetime	No	Timestamp when the review was last updated
bookId	int	Yes	Foreign key reference to the book being reviewed
buyerId	int	Yes	Foreign key reference to the user who wrote the review

Chapter-VI

System Workflow and Security Structure

This chapter describes the major system modules, the internal workflow of processes, important screens with their functionality, and the security measures implemented in the **Second Hand Book Selling** system to ensure safe, efficient, and user-friendly operations.

6.1 Key Modules

The system is divided into the following main modules:

User Module

Handles user registration, login, password encryption, profile management, and password reset using security questions.

Book Module

Enables verified users to list new books for sale, edit their listings, delete listings, and view detailed information about books.

Offer Module

Allows buyers to place offers on listed books, view negotiation history, and sellers to accept, reject, or finalize deals.

Admin Module

Provides the admin with the ability to verify users after registration, ensuring only legitimate users can access sensitive system functionalities.

Review Module

After the completion of a transaction, buyers can leave feedback and ratings for the sellers, promoting trust within the community.

6.2 Process Logic / Workflows

The core workflows of the system are as follows:

User Registration and Verification

A student registers by providing personal details. Admin verifies the account before enabling full system access.

Login and Profile Access

Verified users log in securely using encrypted credentials and JWT-based authentication tokens. They can view and update their profile information.

Book Listing and Viewing

Verified sellers can list books with details like title, author, description, and price. Buyers can browse, search, and view book listings.

Offer Management

Buyers can make price offers for listed books. Sellers review incoming offers and choose to accept, reject, or counteroffer. Finalized offers are marked as sold.

Review Submission

After a successful transaction, users submit reviews and ratings to promote transparent and trustworthy deals.

6.3 Screens and Their Functions

Screen Name	Functionality
Login/Register Screen	Handles user registration and authentication.
Profile Page	Displays user details, allows editing, and provides access to book listings and offer management.
Book Listing Screen (Shop Page)	Enables sellers to upload books with details and buyers to browse available books.

Product Page (Book Details Page)	Displays detailed information about a selected book and provides an option for the buyer to make an offer to the seller.
Offer Management Screen	Allows sellers to manage received offers, and buyers to track sent offers.
Admin Dashboard	Enables the admin to verify user accounts and monitor activities.
Password Reset Screen	Allows users to reset forgotten passwords using security questions.
Review Screen	Lets buyers leave feedback and ratings after a deal is completed.

6.4 Security Implementation

The system ensures data privacy and transaction security through the following measures:

JWT (JSON Web Token) Authentication

Secure tokens are generated during login to maintain authenticated user sessions and protect private routes.

Password Encryption (bcrypt)

User passwords are encrypted using bcrypt before storing them in the database, preventing exposure of raw passwords even if data is compromised.

Security Questions for Password Reset

An additional security layer is provided through custom security questions and answers for password recovery.

Admin Verification

All users must be manually verified by the admin to access offer-making and profile visibility features, ensuring only genuine users interact on the platform.

Chapter-VII

Testing

Testing is an essential phase in the software development lifecycle to ensure that the system meets its intended requirements, functions correctly, and is free of critical bugs. The **Second Hand Book Selling System** underwent several types of testing to validate both frontend and backend functionalities.

7.1 Unit Testing

Unit testing was performed on individual components and functions of the system. Each unit, such as user authentication, book listing, offer handling, and profile management, was tested separately to ensure that they worked independently without errors.

Unit Test Case 1: Login Test

Test Case	Login Test
Item Tested	User login with incorrect credentials
Sample Input	Entered wrong email or password
Expected Output	"Invalid email or password" message
Actual Output	"Invalid email or password"
Result	Successful

Unit Test Case 2: New User Registration

Test Case	Registration Test
Item Tested	Registration with invalid input
Sample Input	Missing email or short password
Expected Output	Error message for missing/invalid input
Actual Output	Error shown as expected
Result	Successful

Unit Test Case 3: Book Listing

Test Case	Book Listing Test
Item Tested	Book form with incomplete details
Sample Input	Left title/price field blank
Expected Output	"All fields required" warning
Actual Output	Same as expected
Result	Successful

Unit Test Case 4: Offer Creation

Test Case	Offer Creation Test
Item Tested	Buyer makes an offer
Sample Input	Offered ₹400 for a book listed at ₹500
Expected Output	"Offer submitted to seller"
Actual Output	Same as expected
Result	Successful

Unit Test Case 5: Mark as Sold

Test Case	Mark as Sold Test
Item Tested	Seller completes deal
Sample Input	Selected offer → entered final price ₹450
Expected Output	Book marked as sold with final price shown
Actual Output	Same as expected
Result	Successful

Unit Test Case 6: Review Submission

Test Case	Review Submission Test
Item Tested	Submitting a review after deal completion
Sample Input	Submitted 5-star rating + review text
Expected Output	"Review added successfully"
Actual Output	Same as expected
Result	Successful

All individual units passed the tests and behaved as expected.

7.2 Functional Testing

Functional testing ensured that the complete system performed the required operations according to the business requirements.

Feature	Test Performed	Purpose	Result
User Registration	Tested user registration with valid and invalid data	To ensure users can successfully create accounts	Pass
User Login	Entered correct and incorrect login credentials	To verify secure authentication and error handling	Pass

Book Listing	Verified book upload, edit, and delete functionalities	To confirm sellers can manage book listings	Pass
Offer System	Tested sending, accepting, and rejecting offers	To ensure buyers and sellers can negotiate securely	Pass
Mark as Sold	Simulated marking a book as sold with final price entry	To confirm transaction finalization flow	Pass
Review Submission	Tested review submission after deal completion	To verify only eligible users can leave feedback	Pass
Admin Verification	Verified admin workflow for verifying new user accounts	To ensure that only approved users gain access to platform	Pass
Password Reset	Answered security question and reset password	To confirm secure password recovery process	Pass

Functional testing confirmed that the system workflows were correct and consistent.

Chapter-VIII

Limitations, future enhancements and conclusion

8.1 System Limitations

While the **Second Hand Book Selling System** fulfills its intended objectives effectively, there are a few limitations:

No Payment Integration:

The platform facilitates offers and negotiations but does not handle online payment or money transactions.

Limited Admin Controls:

Admins currently verify users manually but cannot manage book listings or handle disputes within the platform.

No Automated Notifications:

Users do not receive automatic email or SMS notifications for important activities like offer acceptance or password resets.

8.2 Future Enhancements

Several features can be considered for future upgrades to improve system usability and performance:

Online Payment Integration:

Implementing secure online payment gateways to handle book purchases directly on the platform.

Automated Email/SMS Notifications:

Adding notification services to inform users about important activities like offers, verifications, and sales.

Admin Control Panel Upgrade:

Providing admin with additional options to manage book listings, monitor reviews, and handle disputes.

Recommendation System:

Suggesting books to users based on their browsing history and interests using machine learning.

Mobile Application Development:

Creating a lightweight mobile application to make access easier and more flexible for students.

8.3 Conclusion

The **Second Hand Book Selling System** successfully provides a platform for students to exchange academic resources efficiently and securely. The system addresses key challenges such as the lack of verified users, absence of structured negotiation, and privacy concerns present in traditional methods.

By implementing a centralized, secure, and user-friendly solution, the platform encourages book reuse, promotes sustainability, and helps students save money.

Although there are a few limitations, the foundation laid by this system offers significant potential for future expansion and enhancement.

Overall, the project has achieved its primary objectives and demonstrates practical application of web development concepts, database management, and system design principles.

References

Books:

- **"Database System Concepts"** — *Abraham Silberschatz, Henry F. Korth, and S. Sudarshan*
- **"Software Engineering: A Practitioner's Approach"** — *Roger S. Pressman*

Portals:

- ☐ <https://react.dev/>
- ☐ <https://nodejs.org/en/docs>
- ☐ <https://expressjs.com/>
- ☐ <https://dev.mysql.com/doc/>
- ☐ <https://mui.com/>
- ☐ <https://www.postman.com/>
- ☐ <https://github.com/kelektiv/node.bcrypt.js>
- ☐ <https://jwt.io/introduction>
- ☐ <https://www.youtube.com/>

USER GUIDE

This user manual provides a step-by-step guide on how to use the **Second-Hand Book Selling System**, including both student users and admin roles.

1. Getting Started

For Student Users (Buyers & Sellers)

1.1 Registration

- Navigate to the **Register** page.
- Fill in your personal details including name, email, password, contact, and security question.
- Submit the form.
- Wait for **admin verification** before gaining full access.

1.2 Login

- Go to the **Login** page.
- Enter your registered email and password.
- If verified by the admin, you will be redirected to your profile page.

2. User Dashboard / Profile

Once logged in, users can:

- View and **edit their profile information**.
- Access:
 - **Manage Offers** (received or sent)
 - **View Listed Books**
 - **Sell a Book**

3. Book Selling Process

3.1 Listing a Book

- Navigate to **Sell a Book** from the profile.
- Fill in book details (title, author, price, description, image).
- Submit the form to list it on the platform.

3.2 Managing Book Listings

- Use **View Book Listing** to edit or delete your listed books.

4. Book Buying / Offer System

4.1 Browsing Books

- Go to the **Shop Page**.
- Browse or search for listed books.
- Click on a book to open the **Product Page**.

4.2 Making an Offer

- On the Product Page, enter an offer price.
- Submit the offer to the seller.
- Await seller's response (Accept / Reject).

5. Offer Management (Seller Side)

- From the **Manage Offers** section, view all incoming offers.
- You can:
 - **Accept** or **Reject** any offer.
 - Once accepted, **Mark as Sold** to finalize the deal.

5.2 Finalizing Deal

- When marking a book as sold, enter the final agreed price.
- The book status will update to **Sold**.

6. Review System

- After a deal is marked as sold:
 - Buyer can leave a **rating and review**.
 - Reviews help maintain trust and platform quality.

7. Password Recovery

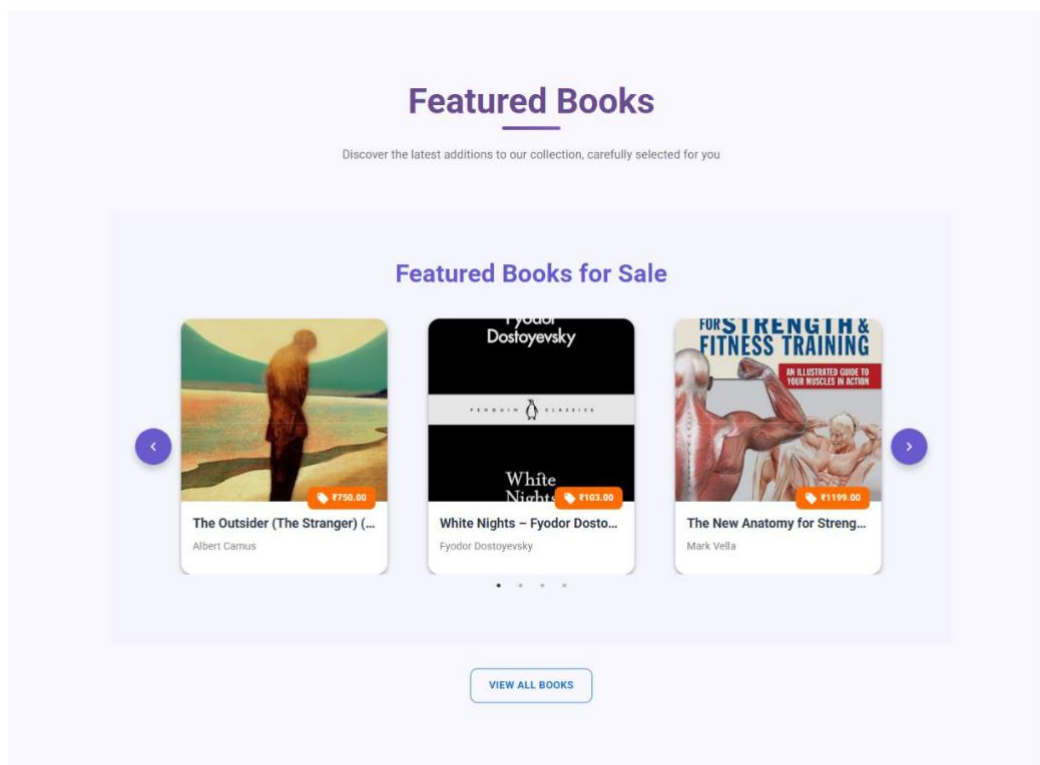
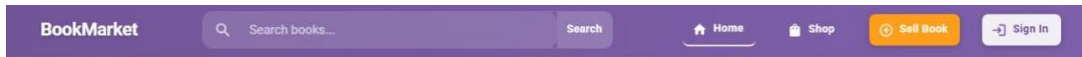
- If you forget your password:
 - Click on **Forgot Password** on the login page.
 - Enter your email and answer the **security question** set during registration.
 - Reset your password securely.

8. Admin User

Guide Admin Functions

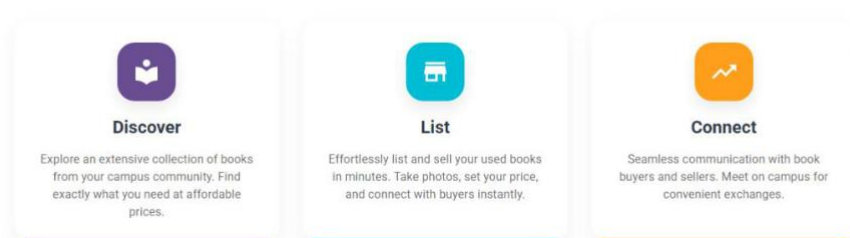
- Log in to the admin panel (secured).
- **Verify new users** to enable them to access the system.
- Monitor platform activity.

SCREENSHOTS



How BookMarket Works

Your campus book exchange reimagined for the digital age



Why BookMarket?

Join thousands of students who are transforming how textbooks are bought and sold



Save Money on Expensive Books

College textbooks, especially technical and professional ones, can be costly. Second-hand books are much more affordable.



Eco-Friendly Option

Reusing books reduces waste and supports sustainable habits—small steps towards a greener future.



Supports Rural and Urban Students Alike

Whether you are in a big city or a small town, the platform connects you with the right book at the right price.



Earn by Selling Old Books

After completing a semester, sell unused books and earn pocket money instead of letting them pile up.



BookMarket**Search**[Home](#)[Shop](#)[Sell Book](#)[Sign In](#)

Ready to Join the Community?

Start buying and selling books with your campus community today

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Quick Links

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Fig : 1 (Home page)

Book Collection

Discover your next favorite book from our curated selection

Filters

Book Condition

Min Price

Max Price

Apply Filters

Reset All

Sort Options

Sort by

Sort direction

Grow Your Own Medicine: Ha...

Ava Green & Kate Bensinger

Price: ₹4500

Condition: Good

Genre: Biology

VIEW BOOK DETAILS

The Herb Book (Hardcover)

DK

Price: ₹501

Condition: Fair

Genre: Biology

VIEW BOOK DETAILS

Patternmaking for Fashion D...

Armstrong

Price: ₹845

Condition: Poor

Genre: Fashion

VIEW BOOK DETAILS

End Of Fashion (Paperback)

Teri Agins

Price: ₹542

Condition: Good

Genre: Fashion

VIEW BOOK DETAILS

Songs of Salvation: By Dora ...

Dora Greenwell

Price: ₹370

Condition: Poor

Genre: Poems

VIEW BOOK DETAILS

Selected Poems From The W...

Dora Greenwell

Price: ₹563

Condition: Fair

Genre: Poe

VIEW BOOK DETAILS

1

2

Fig : 2(Shop Page)

47



BookMarket
Buy and sell textbooks with fellow students

Username * Email *

Password *

Full Name * Phone Number *
Phone number must be 10 digits

Bio (optional) Location (optional)

Security Questions (Required)

CREATE ACCOUNT

Already have an account? [Log in here](#)

Fig 3 : Create New Account



BookMarket
Welcome back! Log in to your account.

Email

Password

[Forgot password?](#)

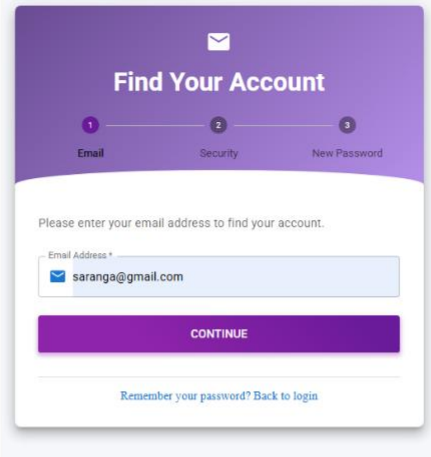
LOG IN

New to BookMarket?

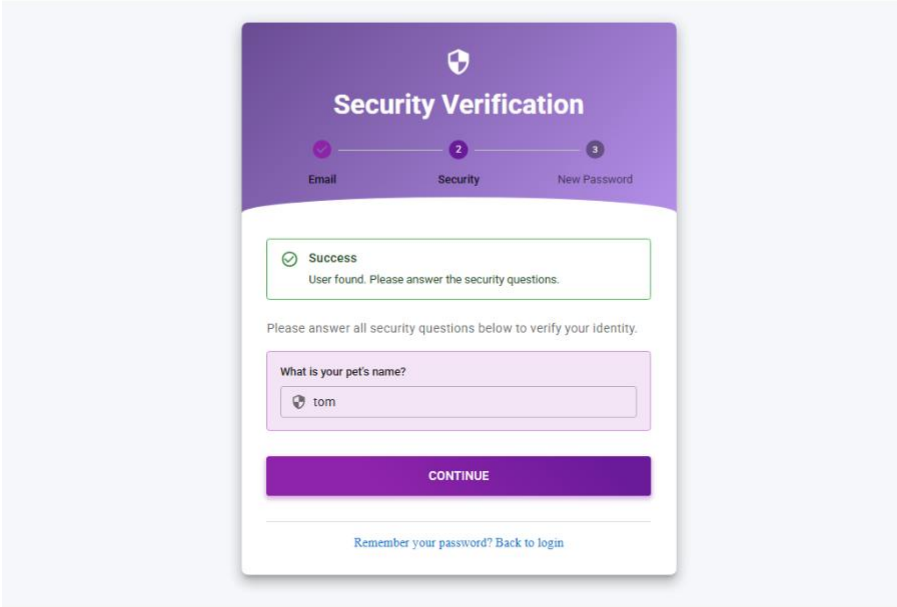
[CREATE NEW ACCOUNT](#)

By logging in, you agree to CollegeBookHub's Terms of Service and Privacy Policy

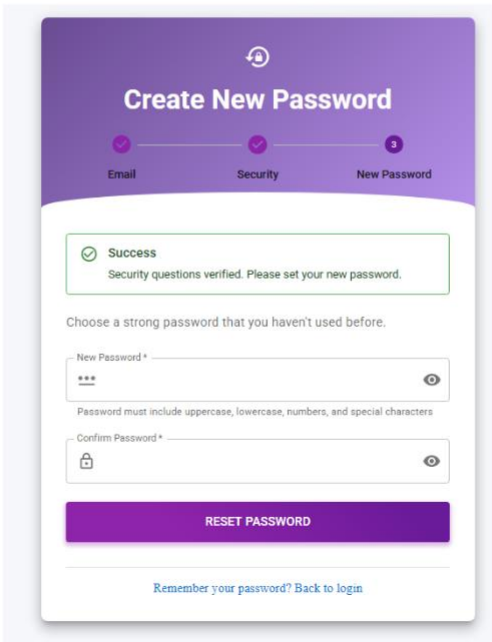
Fig 4 : Login



The 'Find Your Account' screen features a purple header with an envelope icon and the title 'Find Your Account'. A progress bar below the header shows three steps: '1 Email' (active), '2 Security', and '3 New Password'. The main content area is white and contains the instruction 'Please enter your email address to find your account.' Below this is a text input field labeled 'Email Address *' with the email 'saranga@gmail.com' entered. A purple 'CONTINUE' button is positioned below the input field. At the bottom, there is a link that reads 'Remember your password? Back to login'.



The 'Security Verification' screen has a purple header with a shield icon and the title 'Security Verification'. The progress bar shows '1 Email' as completed, '2 Security' as the current step, and '3 New Password' as the next step. A green success message box states 'Success User found. Please answer the security questions.' Below this, the instruction 'Please answer all security questions below to verify your identity.' is displayed. A text input field is labeled 'What is your pet's name?' and contains the text 'tom'. A purple 'CONTINUE' button is located below the input field. The bottom of the screen features the link 'Remember your password? Back to login'.



The 'Create New Password' screen features a purple header with a circular arrow icon and the title 'Create New Password'. The progress bar shows '1 Email' and '2 Security' as completed steps, and '3 New Password' as the current step. A green success message box indicates 'Success Security questions verified. Please set your new password.' Below this, the instruction 'Choose a strong password that you haven't used before.' is shown. There are two text input fields: 'New Password *' (with masked characters '***') and 'Confirm Password *' (with a lock icon). A note below the first field states 'Password must include uppercase, lowercase, numbers, and special characters'. A purple 'RESET PASSWORD' button is positioned below the input fields. The bottom of the screen includes the link 'Remember your password? Back to login'.

Fig :5 Reset Password Feature Page



saranga

casually sometimes read out of peer pressure

Member

EDIT

Contact Information

Email

saranga@gmail.com

Phone

4444444444

Location

Dima Hasao

Quick Actions

VIEW MY BOOKS

VIEW OFFERS

LIST NEW BOOK

Fig : 6 Profile Page

Become a Seller

Book Title *

Author *

Year *

ISBN *

Book Type *

Cover Damage?

☐ Yes
☒ No

Marks on Pages?

☐ Yes
☒ No

Binding Intact?

☒ Yes
☐ No

Water Damage?

☐ Yes
☒ No

Missing Pages?

☐ Yes
☒ No

Spine Wear?

☐ Yes
☒ No

Calculated Condition

Good

Quantity Available

1

Price *

Additional Info

Seller Name

saranga

Email

saranga@gmail.com

Phone

4444444444

Address

Dima Hasao.

UPLOAD BOOK IMAGES

Max 3 images (jpg, png, jpeg)

LIST BOOK

Please upload at least one book image before listing

Fig 7 : Selling Page

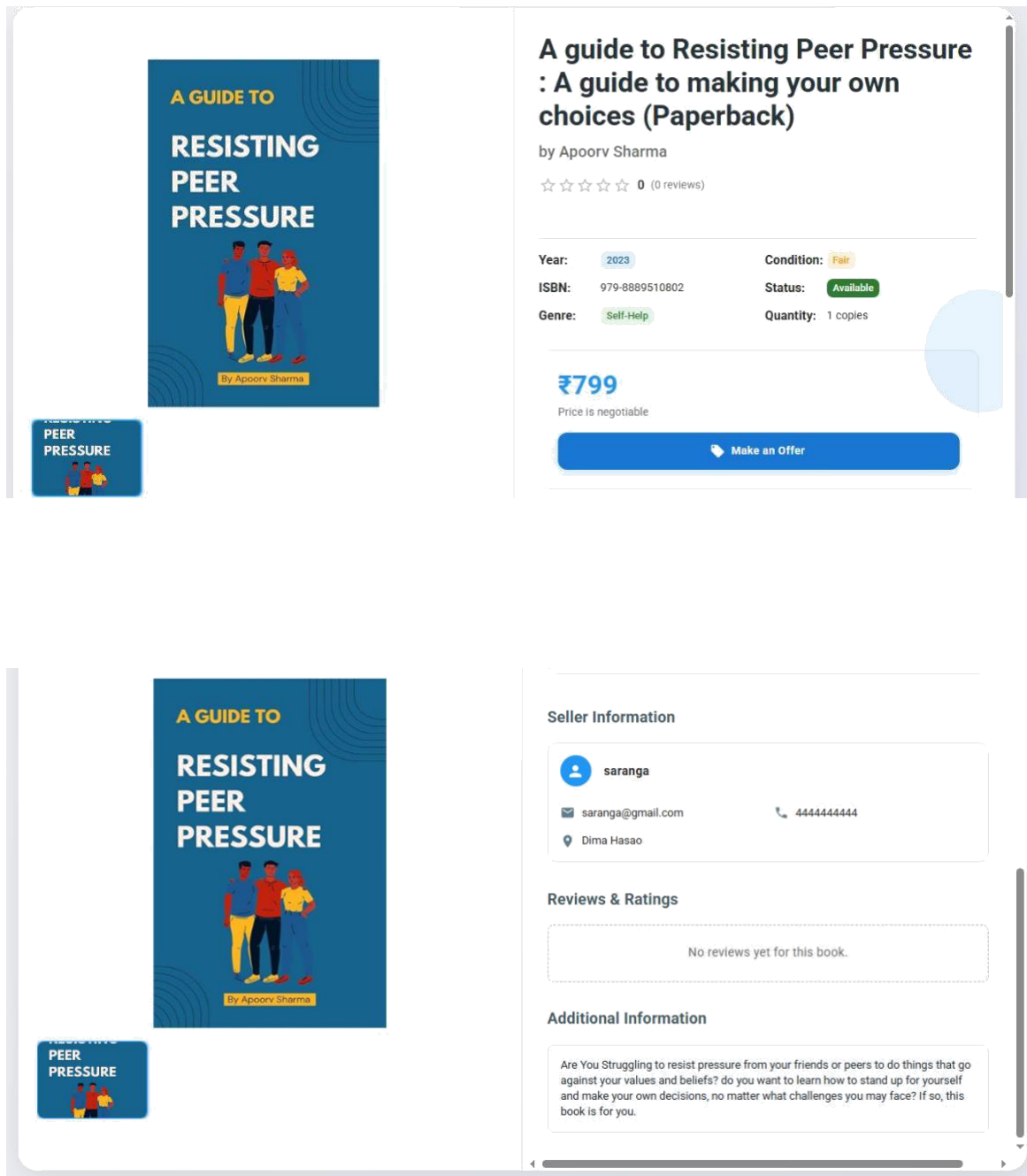


Fig 8 : Product Page

Make an Offer

Current price: ₹799. Enter your offer price below:

Cancel

Submit Offer

Fig 9: Make a Offer Page

Manage Your Offers

YOUR OFFERS (BUYER)

SELLING OFFERS (SELLER)

COMPLETED OFFERS

Book Title	Final Price	Buyer	Seller	Review
^ Songs of Salvation: By Dora Greenwell (Paperback)	₹230	You	naba	VIEW REVIEW
Book Details Author: Dora Greenwell Condition: Poor Original Price: ₹370		Transaction Details Completed On: 26/4/2025 Buyer: You Seller: naba		
Your Review ★★★★☆ i thought it was about dora the explorer REMOVE REVIEW				

Manage Your Offers

YOUR OFFERS (BUYER)		SELLING OFFERS (SELLER)		COMPLETED OFFERS	
Book Title		Offered Price	Status	Actions	
▼	A guide to Resisting Peer Pressure : A guide to making your own choices (Paperback)	₹500	Pending	✓ ACCEPT	✗ DENY

Manage Your Offers

YOUR OFFERS (BUYER)		SELLING OFFERS (SELLER)		COMPLETED OFFERS	
Book Title		Offered Price	Status	Actions	
▼	The Outsider (The Stranger) (Perfect Paperback)	₹100	Pending	✕ CANCEL	

Fig: 10 Manage Your Offer

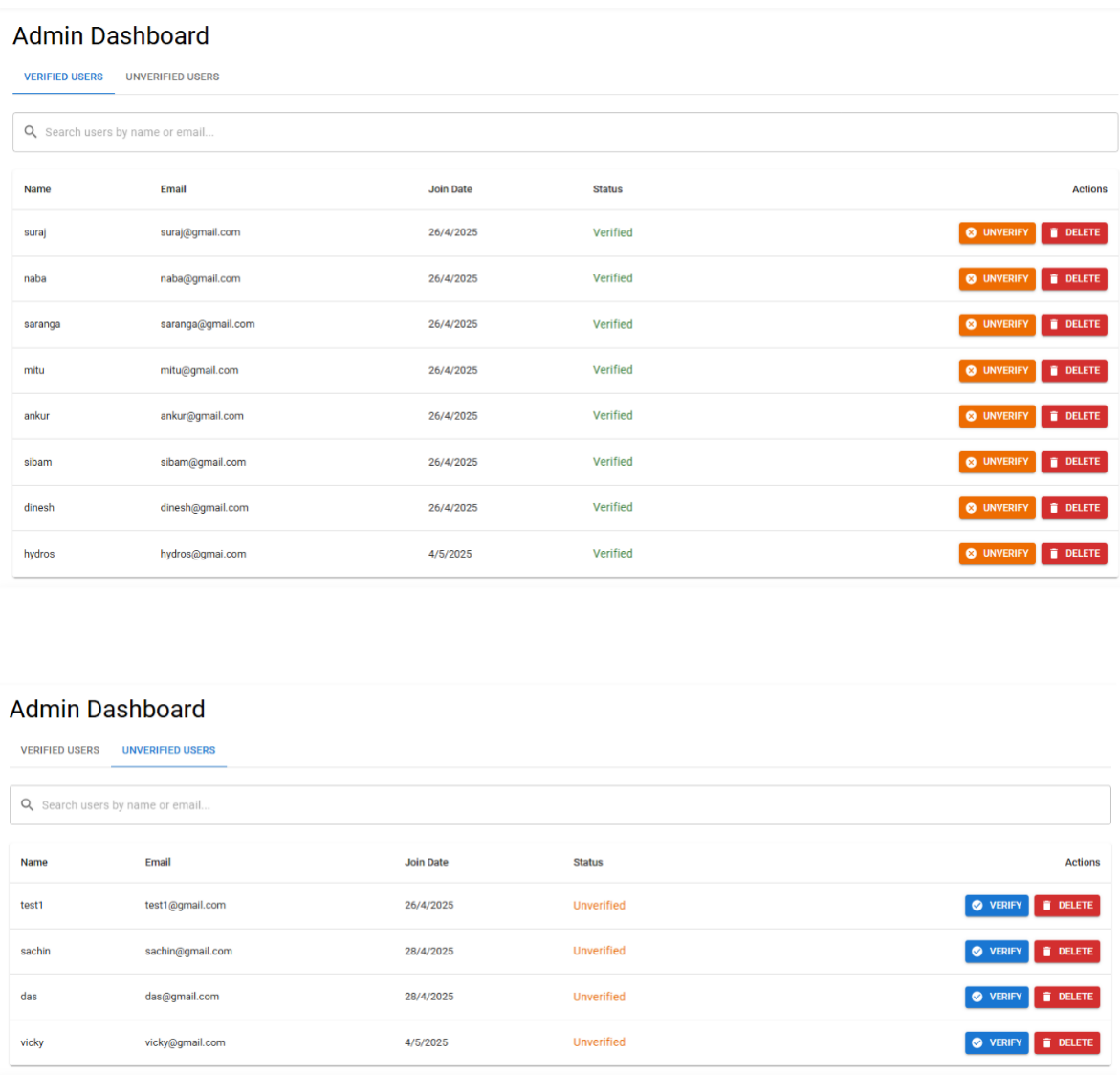


Fig: 11 Admin Page